

ASSIGNMENT N.2 STRENGTHENING OF ALLOYS

Given the following table of engineering alloys, select **one alloy grade** for each of the 4 alloy classes (Al-based, Cu-based, Carbon steel-based, Stainless Steel-based, (excluding the Co-based alloy and the Mo-based alloy)), hence, totally four alloy grades, and define for all of them the following attributes:

1. The strengthening methods used for the selected alloy
2. Roles played by main alloying elements for each strengthening method
3. The shapes and processes used to achieve the aimed strength
4. Weldability e method of welding
5. Typical engineering applications (attach figures if available)
6. Give a plot of the tensile properties (if available)
7. Attach to the Assignment file the main used reference sources.
8. Any outstanding property making the alloy unique (if any)
9. Any weakness of the alloy the frequent causes of failure in practice
10. Possible surface treatment to enhance resistance to corrosion (critical media) and wear.

	K, Mpa	n	
Aluminum, 1100-0 Industrial purity	180	0.2	
2024-T4	690	0.16	
5052-0	210	0.13	→
6061-0	205	0.2	
6061-T6	410	0.05	
7075-0	400	0.17	
Brass, 60-39-1 Pb, annealed	800	0.33	
70-30, annealed	895	0.49	
85-15, cold-rolled	580	0.34	←
Bronze (phosphor), annealed	720	0.46	
Cobalt-base alloy, heat-treated	2070	0.5	
Copper, annealed	315	0.54	
Molybdenum, annealed	725	0.13	
Steel, low-carbon annealed	530	0.26	
1045 hot-rolled	965	0.14	
1112 annealed	760	0.19	
1112 cold-rolled	760	0.08	→
4135 annealed	1015	0.17	
4135 cold-rolled	1100	0.14	
4340 annealed	640	0.15	
17-4 P-H annealed Precipitation hardening steel	1200	0.05	→
52100 annealed	1450	0.07	
302 stainless, annealed	1300	0.3	
304 stainless, annealed	1275	0.45	←
410 stainless, annealed	960	0.1	→